WEEK 8 Skill OSSP

1. Create Week 8 workspace

```
suraj@DESKTOP-13F1P77:~/week7_s$ cd ~
suraj@DESKTOP-13F1P77:~$ mkdir week8_s
suraj@DESKTOP-13F1P77:~$ cd ~/week8_s
suraj@DESKTOP-13F1P77:~/week8_s$ pwd
/home/suraj/week8_s
suraj@DESKTOP-13F1P77:~/week8_s$
```

2. Environment Variable Expansion

```
suraj@DESKTOP-13F1P77:~/week8_s$ nano env_expand.c
suraj@DESKTOP-13F1P77:~/week8_s$ gcc env_expand.c -o env_expand
suraj@DESKTOP-13F1P77:~/week8_s$ ./env_expand
Enter variable reference: $USER
USER = suraj
suraj@DESKTOP-13F1P77:~/week8_s$ ./env_expand
Enter variable reference: $HOME
HOME = /home/suraj
suraj@DESKTOP-13F1P77:~/week8_s$ ./env_expand
Enter variable reference: $ABCXYZ
ABCXYZ = <undefined>
suraj@DESKTOP-13F1P77:~/week8_s$
```

3. Replace the previous program and Compile

```
suraj@DESKTOP-13F1P77:~/week8_s$ nano env_expand.c
suraj@DESKTOP-13F1P77:~/week8_s$ gcc env_expand.c -o env_expand
suraj@DESKTOP-13F1P77:~/week8_s$ ./env_expand
Enter variable reference: $USER
Original token : $USER
Expanded token : suraj
suraj@DESKTOP-13F1P77:~/week8_s$
```

4. Test Nested Variable Expansion

```
suraj@DESKTOP-13F1P77:~/week8_s$ export NAME="OSSP"
suraj@DESKTOP-13F1P77:~/week8_s$ export GREETING='$NAME'
suraj@DESKTOP-13F1P77:~/week8_s$ echo $GREETING
$NAME
suraj@DESKTOP-13F1P77:~/week8_s$ ./env_expand
Enter variable reference: $GREETING
Original token : $GREETING
Expanded token : $NAME
suraj@DESKTOP-13F1P77:~/week8_s$ ./env_expand
Enter variable reference: $NAME
Original token : $NAME
Expanded token : OSSP
suraj@DESKTOP-13F1P77:~/week8_s$
```

5. Built-in Command Dispatch

```
suraj@DESKTOP-13F1P77:~/week8_s$ nano builtin.c
suraj@DESKTOP-13F1P77:~/week8_s$ nano builtin.c
suraj@DESKTOP-13F1P77:~/week8_s$ gcc builtin.c -o builtin
suraj@DESKTOP-13F1P77:~/week8_s$ ./builtin
mini-shell> pwd
/home/suraj/week8_s
mini-shell> hello
Invalid command: hello
mini-shell> pwd
/home/suraj/week8_s
mini-shell>
```

6. Test Exit Built-in

```
mini-shell> exit
Exiting shell...
suraj@DESKTOP-13F1P77:~/week8_s$
```

Summary:

Implemented environment variable expansion during command parsing in Ubuntu. The program detected $VAR references, expanded their values using environment variables, handled undefined variables, updated tokens, and demonstrated nested variable expansion. Implemented built-in command dispatch using a dispatch table. Built-in commands such as pwd and exit were executed within the shell, invalid commands were detected, and shell state was maintained and tested successfully.